

Beyond the Proxy: Trajectory-Distilled Guidance for Offline GFlowNet Training

Proxy-free training via IRL edge rewards — indirect guidance, ground-truth gradients.

Ruishuo Chen · Xun Wang · Rui Hu · Zhuoran Li · Longbo Huang✉

Institute for Interdisciplinary Information Sciences · Tsinghua University · Beijing, China

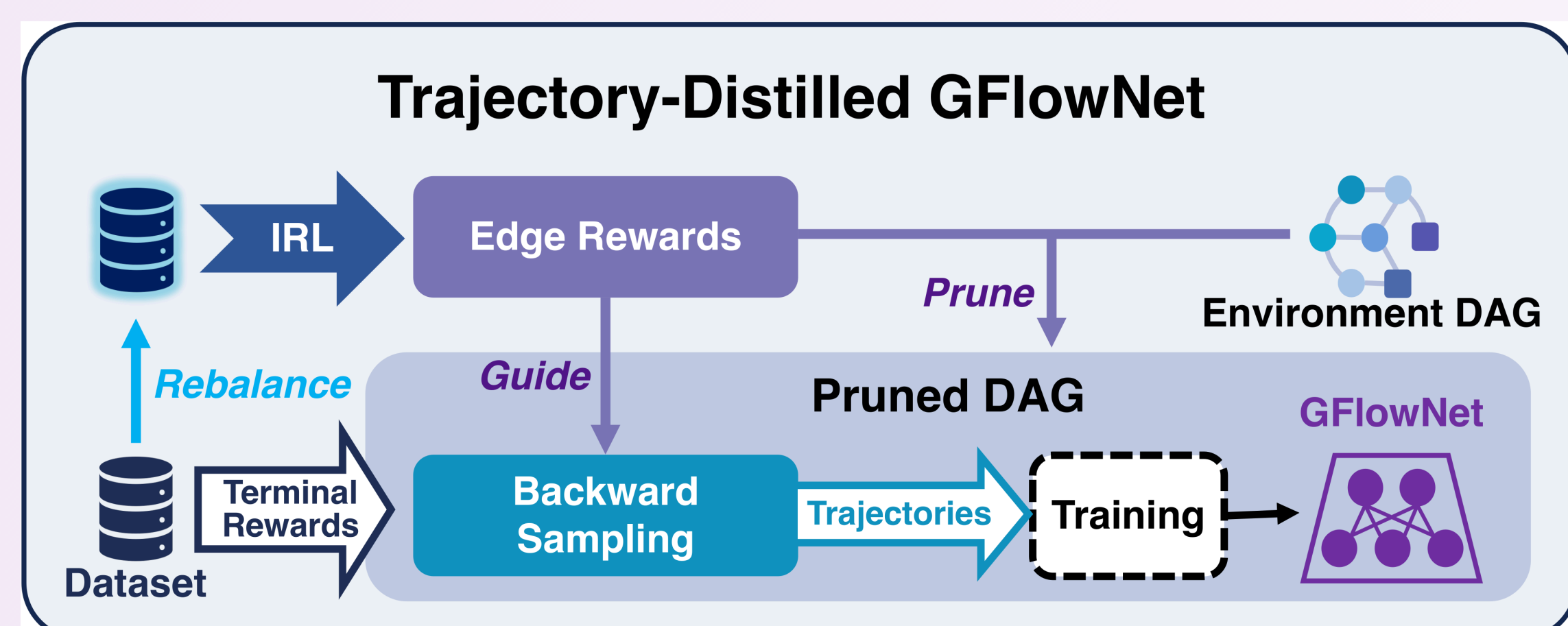


清华大学 交叉信息研究院

Institute for Interdisciplinary Information Sciences, Tsinghua University



Code & Models



FRAMEWORK TD-GFN distills **edge rewards** from offline trajectories via IRL on a rebalanced dataset, then uses them *indirectly* — **DAG pruning** + **prioritized backward sampling** — so gradients stay on ground-truth terminal rewards.

6×
faster mode discovery
(Hypergrid 8⁴)

16×
fewer trajectories
vs Oracle (Molecule)

SOTA
3 benchmarks:
Grid · AMP · Mol

Proxy-free
no OOD reward queries;
error-propagation-safe

1 Motivation

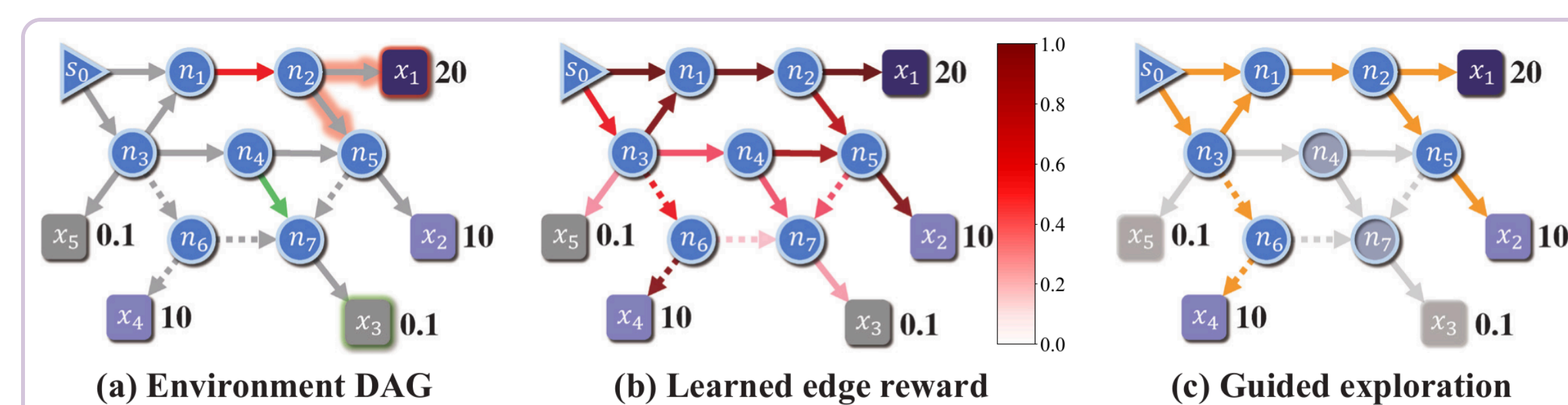
GFlowNets sample diverse high-reward objects, but in many domains active reward queries are **infeasible** (wet labs, human eval). Standard offline pipelines depend on a learned **reward proxy**:

- Reliable proxy needs *large, diverse* data — rarely available.
- OOD queries leak *fitting error* into the policy gradient.
- Existing proxy-free methods impose *coarse* constraints.

Q: Use offline trajectories — not proxies — to give the policy *dense, transition-level* guidance?

2 Key Insight: Edges Differ

DAG transitions contribute **unequally** to policy learning. Removing one edge can block a reward-20 mode; another only loses a 0.1 leaf.



(a) environment DAG — solid: observed, dashed: unobserved. (b) edge rewards R_E recovered by IRL. (c) guided exploration over the pruned DAG.

Implication. Quantify edge importance offline → steer exploration *without* a proxy — foundation of TD-GFN.

3 Phase 1: IRL Edge Rewards

Exploit the GFlowNet-entropy-RL equivalence: train discriminator D_ϕ on a **rebalanced** dataset $\tilde{D}$ ($P(\tau) \propto R(s_\tau)$) via max-causal-entropy IRL:

EDGE REWARD (UNBIASED; KOSTRIKOV ET AL. 2018)

$$R_E(s, s') = \log D_\phi(s, s') - \log(1 - D_\phi(s, s'))$$

Rebalancing biases sampling toward high-reward terminals, yielding a *pseudo-expert* set whose backward policy is still $\mathcal{P}_B^*$ (Prop. 1).

Thm 1 (Recovery + Stability). Among parent-normalised global minimizers of the population IRL objective, $r^* = \log \mathcal{P}_B^*$ is unique. If $\|R_E - r^*\|_\infty \leq \varepsilon$, then $\text{TV}(\hat{\mathcal{P}}_B, \mathcal{P}_B^*) \leq \frac{1}{2}(e^{2\varepsilon} - 1)$.

4 Phases 2-3: Indirect Guidance ★ KEY

Edge rewards *never* enter the gradient. Two indirect mechanisms shape training:

(I) **PRUNE LOW-UTILITY EDGES** · (II) **BACKWARD-SAMPLE ON PRUNED $G' = (V', E')$**

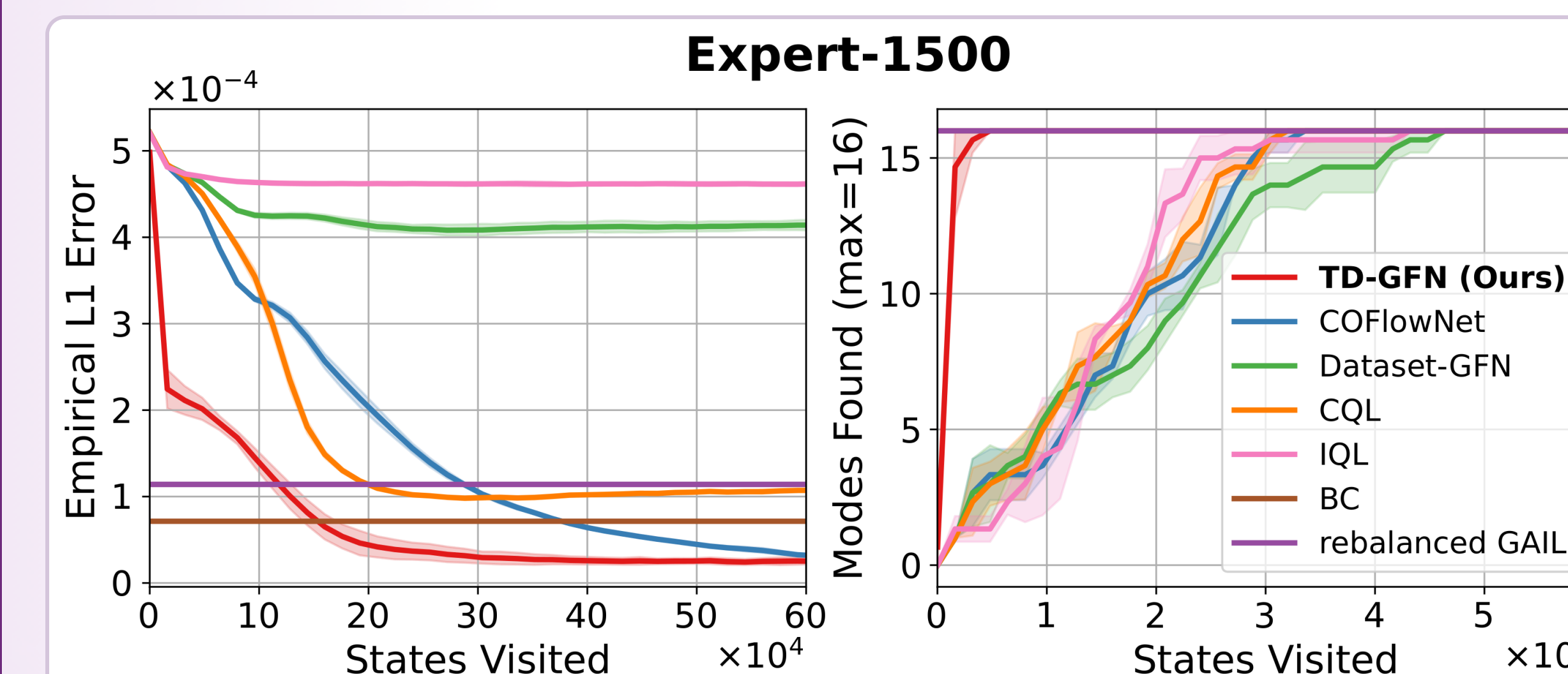
$$\text{drop}(s, s') \text{ if } R_E < \text{mean} - K \cdot \text{std}; \quad \mathcal{P}_B(s_t | s_{t+1}) \propto_{E'} e^{R_E(s_t, s_{t+1})}$$

Policy is optimised on G' using **only ground-truth terminal rewards** — **no error propagation**, while backward sampling concentrates effort on high-utility paths.

Thm 2 (Robust Pruning). If $\|R_E - r^*\|_\infty \leq \varepsilon$, every terminal with bottleneck $\beta(x) > \tau + \varepsilon$ survives; on survivors $\mathcal{X}' \neq \emptyset$ the reward-proportional target is preserved exactly up to renormalisation.

5 Hypergrid 8⁴ ★ 6× Faster Modes

1,500 expert traj. · FM objective · vs. COFlowNet, CQL, IQL, BC, rebal. GAIL, Dataset-GFN.



Left: L_1 — 2× faster than COFlowNet. **Right:** all 16 modes in < 5,000 visits — **6×** speed-up.

16/16
modes found
(< 5×10^3 visits)

2×
faster L_1
convergence

6×
faster vs
baselines

6 Robust Across Data Regimes

TD-GFN matches the target distribution and recovers modes *first* under every stress test:

- Mixed** — 50% noise from random policy.
- Few** — only 1/10 trajectories (150).
- Median** — half-trained behavior policy.
- Bad** — inverted-reward behavior policy, $IR = R^{-0.1}$ (worst-case demos).

Even when only **6/16** modes are covered by the offline data, TD-GFN *generates* the remaining modes via R_E generalisation — the policy reaches DAG regions never visited at training time.

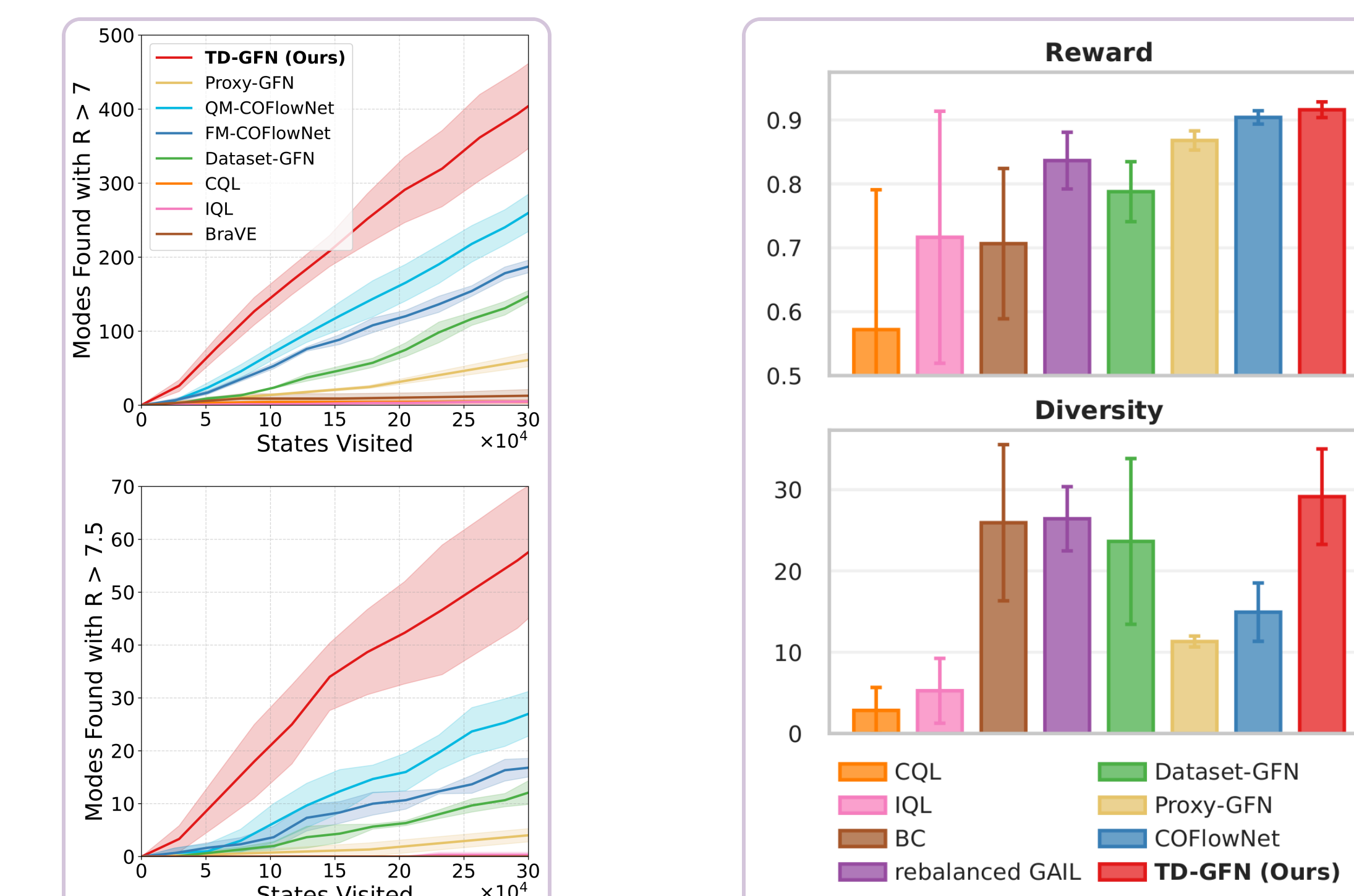
7 Molecule Design ★ SOTA + Oracle-Level

5,000 samples per policy · means over 3 seeds (std in paper) · 1,500-traj. offline set · Conv. = trajectories to convergence.

Method	R-10 ↑	R-100 ↑	Conv. ↓ ($\times 10^4$)
Oracle-GFN (ref.)	7.718	7.408	44.14
Proxy-GFN	7.625	7.281	43.74
FM-COFlowNet	7.582	7.201	5.83
QM-COFlowNet	7.611	7.296	4.42
TD-GFN (Ours)	7.733	7.450	2.75

Top reward among offline baselines; among GFN-style methods uses the **fewest trajectories** — matching Oracle-GFN at **1/16 the cost**.

8 Diversity: Molecule & Peptide



Mol. 1.5–2× more high-reward modes ($R > 7$, $R > 7.5$).

AMP. Top-100 reward & diversity — both best.

Pruning does *not* hurt diversity — TD-GFN beats every offline baseline on **both** tasks.

9 Takeaways

PROXY-FREE. No reward model, no OOD queries, no fitting error in gradients.

DENSE GUIDANCE. IRL distills transition-level R_E from offline trajectories.

INDIRECT & SAFE. Prune + backward-sample; gradients stay on ground-truth rewards.

EFFICIENT SOTA. 6× faster modes, 16 × fewer trajectories than Oracle.